

Somaiya Vidyavihar University
K J Somaiya School of Engineering
Department of Electronics and Telecommunication Engineering



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Batch: A2 Roll No.: 16010324044
Experiment / assignment / tutorial No. 03
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

Sw
12/08/26

Experiment No: 3

Title : Implementation of Database in SQL -DDL

Objective: Define/modify database definitions with proper constraints

OUTCOME:

CO3. Understand the internal storage structures in a physical DB design

Books/ Journals/ Websites referred:

1. G. K. Gupta : "Database Management Systems", McGraw – Hill
2. Korth, Silberchatz, Sudarshan : "Database Systems Concept", 6th Edition , McGraw Hill
3. Elmasri and Navathe, "Fundamentals of Database Systems", 5th Edition, PEARSON Education.

Theory

Theory: The set of relations in a database must be specified to the system by means of a data definition language (DDL). The SQL DDL allows specification of not only a set of relations but also specific information about the relation including,

1. The schema for each relation
2. The domain of values associated with each attribute
3. The integrity constraints
4. The set of indices to be maintained for each relation
5. The security and authorization information for each relation
6. The physical storage structure of each relation on disk

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Syntax Create Table:

```
create table employee(ssn, fname varchar(10), mname varchar(10), lname varchar(10), desg  
varchar(20), gender varchar(5), addr varchar(20), bdate datetime, sal float, primary key(ssn));
```

```
create table manages(ssn int, dept_code int, start_dt datetime, foreign key(ssn) references  
employee(ssn), foreign key(dept_code) references department, key(ssn, dept_code)  
) on delete set null
```

Data Constraints

Business managers of the organization determine a set of rules that must be applied before the data is stored in the database. The application of such rules on raw data ensures **data integrity**.

Eg:- An employee belonging to the Sales department cannot have a salary higher than Rs. 1000.

An employee has an unique identification number.

Applying Data Constraints

Oracle permits data constraints to be attached to table columns using SQL syntax. Constraints can be attached to table columns using Alter table.

Unique Constraint

Unique Constraint- At column level Syntax

<ColumnName><Datatype>(<size>) UNIQUE

Unique Constraint- At table level

CREATE TABLE<TableName>(

<ColumnName><Datatype>(<size>)

<ColumnName><Datatype>(<size>)

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Ex: CREATE TABLE Client (client_id VARCHAR(30) PRIMARY KEY, name VARCHAR(50), phone
VARCHAR(20) UNIQUE CHECK (LENGTH(phone) <= 10), email VARCHAR(50));

CREATE TABLE Client (client_id VARCHAR(30) PRIMARY KEY, name VARCHAR(50), phone
VARCHAR(20) NOT NULL UNIQUE CHECK (LENGTH(phone) <= 10), email VARCHAR(50));

<Columnname><Datatype>(<size>) UNIQUE(<ColumnName1>,<ColumnName2>);

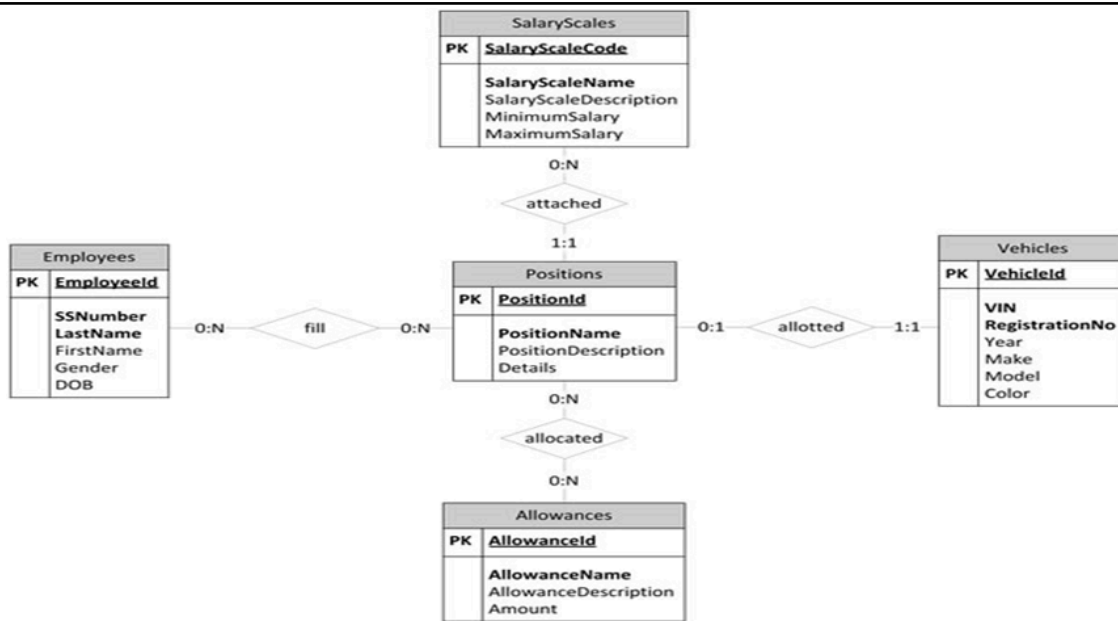
Implementation Details (Problem Statement, Query and Screenshots of Results):

1. For the given ER model, using DDL command: Write syntax to create CREATE Tables with all possible integrity constraints.

Problem Statement:

- 1) A small accounting firm wants a simple HR application that will help it to keep track of its employees, their positions, allowances, salary scales, and which company vehicles their employees drive. The application must keep track of all the positions at the firm, the employees filling these positions, the allowances for these positions, the salary scales for these positions, and the company vehicles assigned to these positions.

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2) For the given ER model, using DDL command: Write syntax to create CREATE Tables with all possible integrity constraints. 1) Put constraint on Phono.no (Client Entity) cant be less than 10 digits and NULL value. 2) Client id should be numeric only.

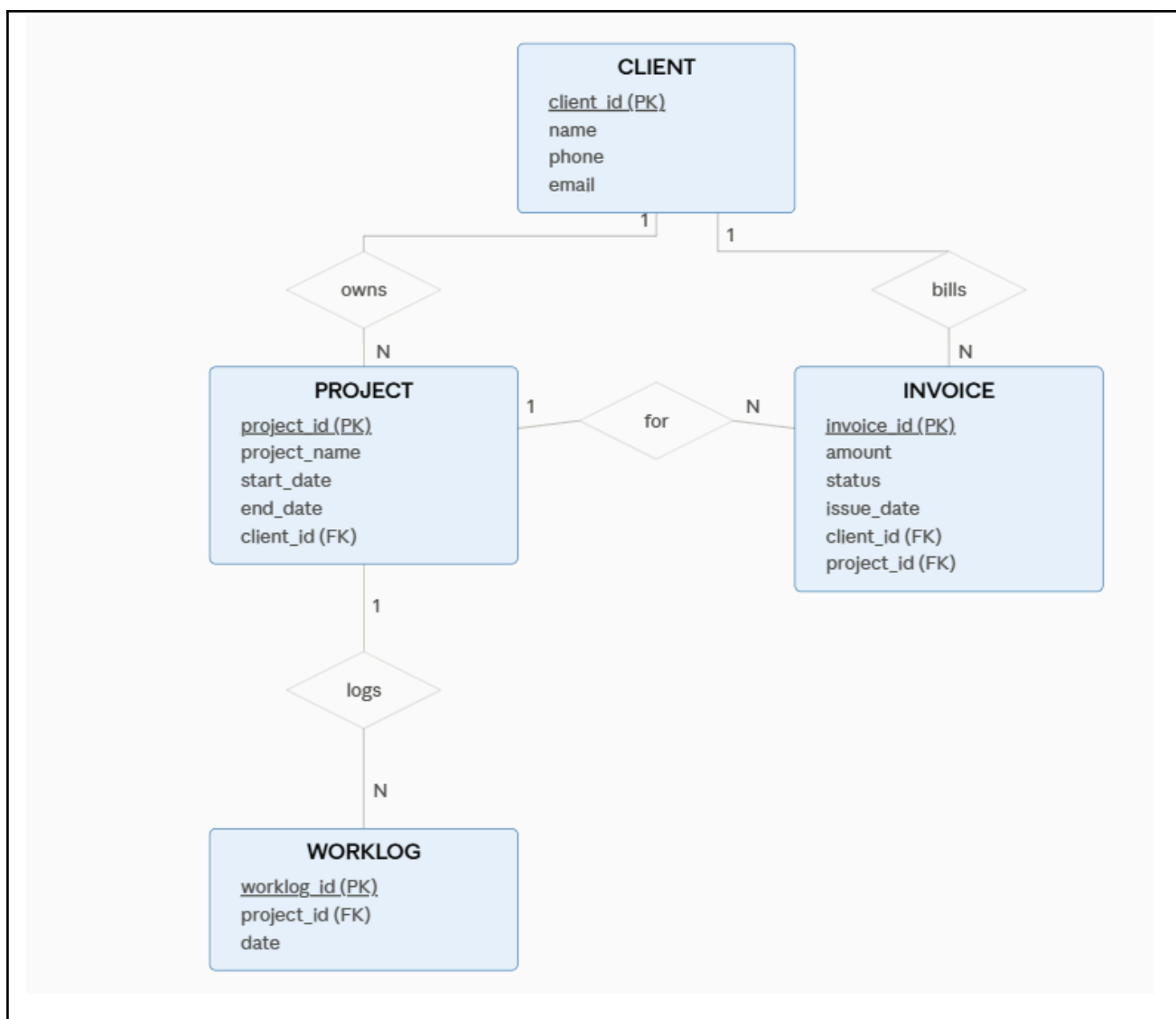
Assigning PK and FK:

```

CREATE TABLE Client (
    client_id VARCHAR(30) PRIMARY KEY,
);

CREATE TABLE Project (
    FOREIGN KEY (client_id) REFERENCES Client(client_id)
);
    
```

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Code:

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Query	Query History
1	CREATE TABLE SalaryScales (
2	SalaryScaleCode INT PRIMARY KEY ,
3	SalaryScaleName VARCHAR(50) NOT NULL ,
4	SalaryScaleDescription VARCHAR(150) ,
5	MinimumSalary NUMERIC(10,2) NOT NULL ,
6	MaximumSalary NUMERIC(10,2) NOT NULL ,
7	
8	CONSTRAINT chk_salary_range
9	CHECK (MinimumSalary <= MaximumSalary),
10	
11	CONSTRAINT chk_min_salary
12	CHECK (MinimumSalary >= 0)
13	);

Query	Query History
1	CREATE TABLE Employees (
2	EmployeeId INT PRIMARY KEY ,
3	SSNumber VARCHAR(20) UNIQUE NOT NULL ,
4	LastName VARCHAR(50) NOT NULL ,
5	FirstName VARCHAR(50) NOT NULL ,
6	Gender VARCHAR(10) ,
7	DOB DATE ,
8	
9	CONSTRAINT chk_gender
10	CHECK (Gender IN ('Male', 'Female', 'Other'))
11	);

Output:

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```

Query  Query History
1  SELECT * FROM salaryscales;

Data Output  Messages  Notifications
+  [Icons]  SQL
salaryscalecode  salaryscalename  salaryscaledescription  minimumsalary  maximumsalary
[PK] integer  character varying (50)  character varying (150)  numeric (10,2)  numeric (10,2)

Total rows: 0  Query complete 00:00:00.145  CRLF  Ln 1, Col 1
  
```

```

Query  Query History
1  SELECT * FROM employees;

Data Output  Messages  Notifications
+  [Icons]  SQL
employeeid  ssnnumber  lastname  firstname  gender  dob
[PK] integer  character varying (20)  character varying (50)  character varying (50)  character varying (10)  date

Total rows: 0  Query complete 00:00:00.152  CRLF  Ln 1, Col 24
  
```

Conclusion:

In this experiment, I successfully created SalaryScales and Employees tables using SQL DDL commands with appropriate data types and integrity constraints. The created tables were then viewed in PostgreSQL (pgAdmin 4), verifying that the database structure was implemented successfully.

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Signature of faculty in-charge

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